AGENTREEL

Process report

Brief C05: AI-ready graduate: how the curriculum and labs build industry AI skills

Team G04 (Group 4): Harshit Modi (16014223039) · Mohammed Zamee Sheikh (16014223054) · Eashan Salian (16014223035)

Run G04-C05-03 · seed 42 · LLM offline · 22 Oct 2026 · L3 · python run.py --brief briefs/C05.yaml --seed 42

51.1s	-14.3	₹0.00	3.4 min	0	20/20
duration (30–60)	LUFS (target –14)	spend of ₹500	wall-clock of 180	interventions of 5	QA rules passed

01 — BRIEF INTERPRETATION & CREATIVE STRATEGY

Show, don't claim

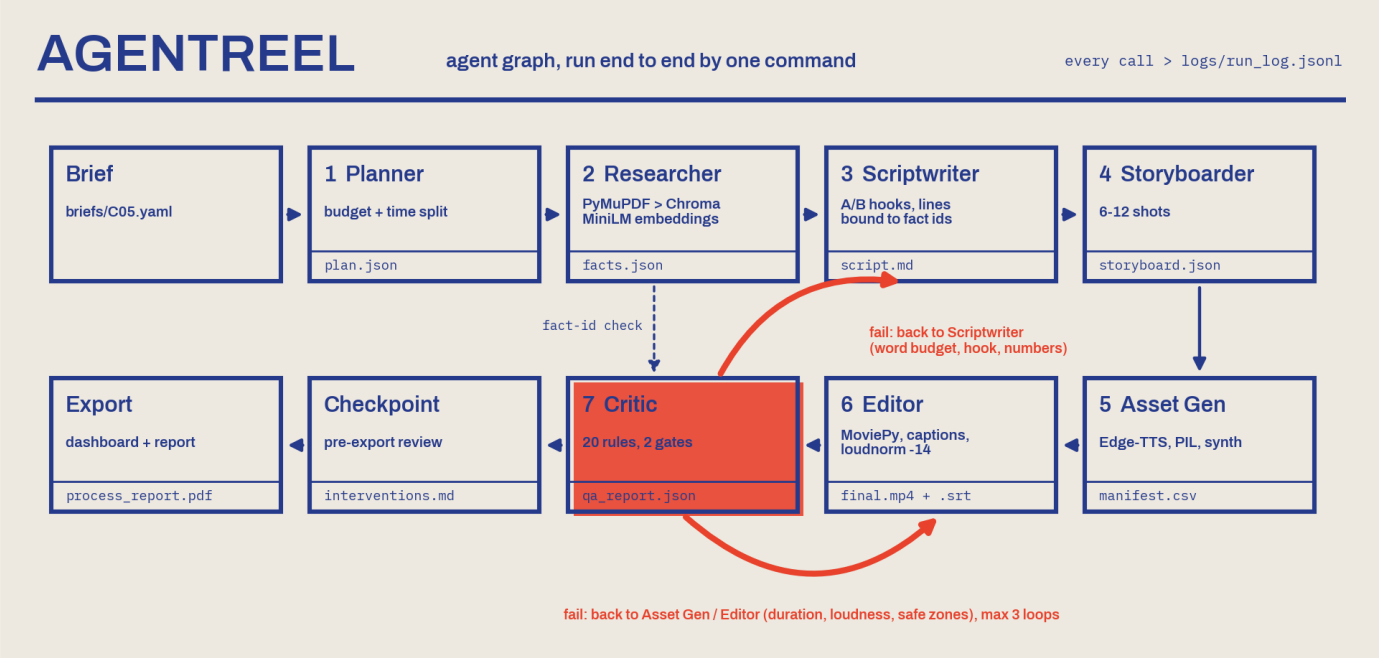
Audience insight. Aspirants want proof the degree turns into real skills; recruiters want evidence graduates can ship, log and govern AI systems. The brief forbids exaggerated placement figures and hallucinated statistics, so the reel sells capability with evidence instead of outcomes: every sentence is traceable to the official LY Agentic AI Lab handout.

Hook (0–3 s). "What does an AI-ready graduate build?" — best of 3 scored variants (score 4.9; others: "What does an AI-ready graduate actually build?" 4.8; "Recruiters, meet the AI-ready graduate." 3.0). **Angle.** Show, don't claim: the reel itself is the proof - built by student-made agents. **CTA.** "Hire our AI-ready graduates today." on a 4 s end card with the AI-disclosure line.

Must include real lab infrastructure (the final-year Agentic AI Lab, local open models, logged and costed runs) and real-world deployment skills (GitHub Pages / Hugging Face Spaces / Streamlit Cloud) — both verified by Critic rules S6/S7.

02 — AGENT GRAPH, FRAMEWORK & MODELS

Seven agents, two critic gates



- **Planner** – turns the brief into plan.json: 9:16 spec, 37.4 s of speech + 4 s end card, 98-word cap, ₹ and minute budgets per stage.
- **Researcher (RAG)** – PyMuPDF → LangChain splitter → ChromaDB with all-MiniLM-L6-v2; returns verbatim claims with source + page.
- **Scriptwriter** – A/B/C hooks scored for the 3-second rule; each body line is bound to fact ids and dropped by priority when over budget.
- **Storyboarder** – 6–12 face-free shots + end card with on-screen text and asset source. **Asset Generator** – Edge-TTS with word timings, procedural visuals, procedural music, manifest.csv.
- **Editor** – MoviePy: Ken Burns, cross-dissolves, karaoke captions in safe zones, ducked music, 2-pass loudnorm. **Critic** – 20 deterministic rules in two gates that route failures back (max 3 loops).

Framework: custom Python orchestrator (run.py) with pydantic schemas on every hand-off. Text agents use a provider-agnostic LLM wrapper (offline seeded template model by default — ₹0 and bit-for-bit reproducible; Claude / OpenAI / Ollama via LLM_PROVIDER). The Critic is intentionally not an LLM.

03 — TIME & COST PER STAGE

₹0.00 spent of the ₹500 cap

STAGE	MODELS / TOOLS	CALLS	TOKENS IN/OUT	TIME (S)	COST ₹
orchestrator		2	0/0	wall	0.00
planner	offline-template	1	310/573	0.0	0.00
researcher	PyMuPDF, chromadb+minilm, offline-template	4	3483/1112	29.7	0.00
scriptwriter	offline-template	2	2468/1681	0.0	0.00
storyboarder	offline-template	1	606/1063	0.0	0.00
asset_generator	PIL/numpy, en-US-AndrewMultilingualNeural, n	20	0/0	71.0	0.00
editor	ffmpeg loudnorm 2-pass, moviepy+libx264	3	0/0	97.2	0.00
critic	rule-engine	4	0/0	0.0	0.00
human		1	0/0	0.0	0.00
Total		**38**		**197.8**	**0.00**

The most expensive stage in time is **editor** (97 s — frame-by-frame PIL compositing at 1080×1920×30 fps). It could be cut by rendering overlays with ffmpeg filters or caching static layers per shot. Paid-API spend is ₹0 because text agents ran on the offline model and TTS/embeddings are free; switching the Scriptwriter to Claude Haiku would cost roughly ₹1–2 per run at the logged token counts.

04 — CRITIC LOOP: WHAT FAILED, WHAT WAS FIXED

Self-repair without a human

- **Script gate · iteration 1 · FAIL** — S1 word budget (127 words vs max 98 (over by 29)) → scriptwriter
- **Script gate · iteration 2 · PASS** — all 8 rules passed
- **Video gate · iteration 1 · PASS** — all 12 rules passed

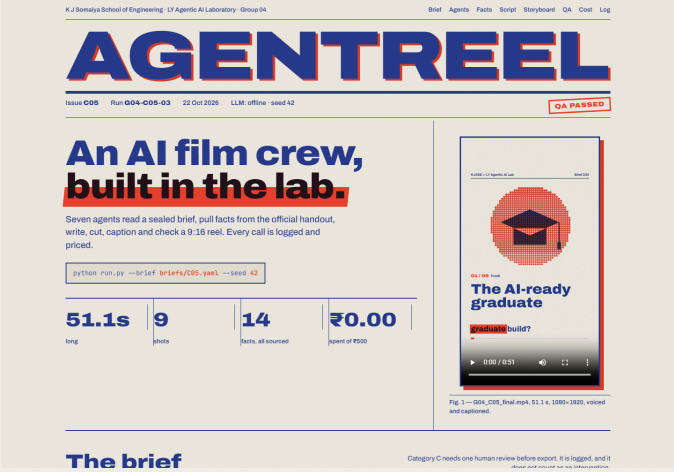
The first draft wrote every grounded line it could (greedy coverage). The Critic measured it against the planner's word budget and routed it back; the Scriptwriter dropped the lowest-priority lines (priority 3–5: retrieval, orchestration, portfolio) while keeping must-include and hook/CTA lines, and v2 passed all eight script rules — no human edit was needed.

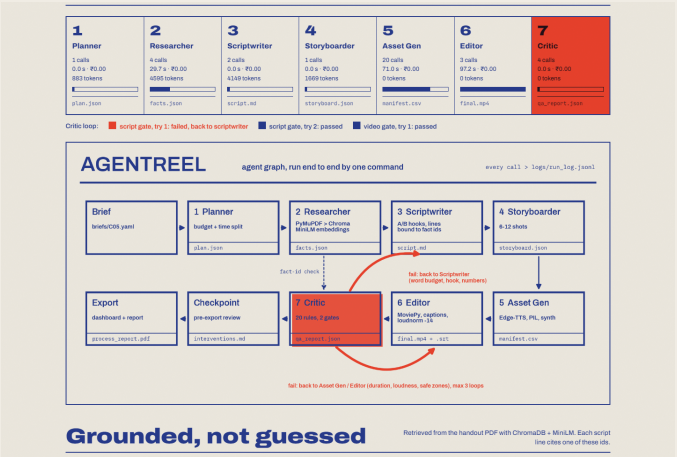
STORYBOARD → RENDERED FRAMES (EDITOR OUTPUT, ONE PER SHOT + END CARD)



05 — FRONTEND & VERIFICATION

The run, auditable at a glance





frontend/index.html renders this run from frontend/data.js: brief card, live agent graph with per-agent calls/time/₹, RAG facts with page citations, scored hooks, storyboard frames, both QA gates, cost table and the raw run log.

Spec check (ffmpeg probe). 1080×1920 · 30 fps · h264 High yuv420p · aac 48000 Hz stereo 164 kb/s · 31.86 MB · -14.3 LUFS · 36 caption cues (Archivo ExtraBold 60 px) · captions y=[1436, 1532] inside the 250/350 px safe zones.

Grounding. 14 facts retrieved (14 verified); every body line cites ≥1 fact id; numbers in the VO (3 hours, ₹500) are matched against facts.json by rule S4.

Submission cut. brag-output/build_reel.py turns this run's timeline, captions, logs and QA report into a HyperFrames (HTML-to-video) composition that shows the real artefacts on screen; it reuses the pipeline's VO + music mix, so the same .srt and the same QA numbers apply.

- √ S1 word budget — 98 words vs max 98
- √ S2 hook <= 3 s — "What does an AI-ready graduate build?" ~2.00s
- √ S3 every body line cites a verified fact — all grounded
- √ S4 no numbers absent from facts.json — numbers ok
- √ S5 must avoid: exaggerated placement figures / stats — clean
- √ S6 must include: real lab infrastructure — mentions the Agentic AI Lab
- √ S7 must include: real-world deployment skills — mentions deployment targets
- √ S8 CTA present — "Hire our AI-ready graduates today."
- √ V1 duration 30-60 s — 51.10s
- √ V2 duration near brief (45s ±20%) — 51.10s vs 45s
- √ V3 1080x1920 (9:16) — 1080x1920
- √ V4 constant 30 fps — 30.0 fps
- √ V5 H.264 High, yuv420p — h264 High yuv420p
- √ V6 AAC 48 kHz stereo 128-192 kbps — aac 48000 Hz stereo 164 kb/s
- √ V7 file size <= 100 MB — 31.86 MB
- √ V8 loudness ≈ -14 LUFS (±1.5) — -14.3 LUFS
- √ V9 burned captions + .srt — 36 cues, outputs/G04_C05_final.srt
- √ V10 text inside safe zones (top 250 / bottom 350 px) — topbar [262, 322], he
- √ V11 hook spoken inside first 3 s — first cue at 0.195s, hook shot ends 2.46s
- √ V12 end card 3-5 s with CTA + AI disclosure — "Created with AI agents — LY A

06 — HUMAN INTERVENTIONS

0 counted · 1 checkpoint

- **22:42 · Critic → export** — Pre-export review checkpoint: approved (Mandatory Category C review (auto-approved (no interactive stdin)); not counted as an intervention)

To remove the checkpoint next time (L4), a second, independent critic (e.g. an LLM safety reviewer voting with the rule engine) would have to sign off Category A/C exports — the handout keeps this review mandatory, so we kept it.

07 — LIMITATIONS, ETHICS & NEXT STEPS

What we would do with another week

- **Fact sheet.** No separate admissions fact sheet was supplied, so the only source is the official lab handout; this deliberately limits claims (no hardware specs, no placement data). The mock-fact fallback is marked unverified and is blocked by rule S3.
- **Visuals.** Official media-kit stills were unavailable; procedural, face-free visuals stand in (drop files into assets/official/ to replace them). No real person's face or voice is generated; the TTS voice is a stock synthetic voice.
- **Licensing & disclosure.** Fonts are SIL OFL, music is generated by the pipeline (CC0), every asset is in assets/manifest.csv; the end card carries "Created with AI agents — LY Agentic AI Lab".
- **Brand.** The end card uses a text wordmark until the official logo is provided (assets/official/logo.png is placed undistorted automatically).
- **Next.** Whisper-aligned captions, Hindi/Marathi auto-dub, multi-critic voting, and a hosted LLM scriptwriter with the same fact-id contract.